As seen in Table 3, SAM uniformly improves performance relative to finetuning without SAM. Furthermore, in many cases, SAM yields novel state-of-the-art performance, including 0.30% error on CIFAR-10, 3.92% error on CIFAR-100, and 11.39% error on ImageNet.

Dataset	EffNet-b7 + SAM	EffNet-b7	Prev. SOTA (ImageNet only)	EffNet-L2 + SAM	EffNet-L2	Prev. SOTA
FGVC_Aircraft	$6.80_{\pm 0.06}$	$8.15_{\pm 0.08}$	**5.3** (TBMSL-Net)	$\mathbf{4.82}_{\pm 0.08}$	$5.80_{\pm 0.1}$	5.3 (TBMSL-Net)
Flowers	$\mathbf{0.63}_{\pm 0.02}$	$1.16_{\pm 0.05}$	0.7 (BiT-M)	$\mathbf{0.35}_{\pm 0.01}$	$0.40_{\pm 0.02}$	0.37 (EffNet)
Oxford_IIIT_Pets	$\mathbf{3.97}_{\pm 0.04}$	$4.24_{\pm 0.09}$	4.1 (Gpipe)	$\mathbf{2.90}_{\pm 0.04}$	$3.08_{\pm 0.04}$	4.1 (Gpipe)
Stanford_Cars	$5.18_{\pm 0.02}$	$5.94_{\pm 0.06}$	**5.0** (TBMSL-Net)	$4.04_{\pm 0.03}$	$4.93_{\pm 0.04}$	**3.8** (DAT)
CIFAR-10	$\mathbf{0.88}_{\pm 0.02}$	$0.95_{\pm 0.03}$	1 (Gpipe)	$\mathbf{0.30}_{\pm 0.01}$	$0.34_{\pm 0.02}$	0.63 (BiT-L)
CIFAR-100	$\mathbf{7.44}_{\pm 0.06}$	$7.68_{\pm 0.06}$	7.83 (BiT-M)	$\mathbf{3.92}_{\pm 0.06}$	$4.07_{\pm 0.08}$	6.49 (BiT-L)
Birdsnap	$\mathbf{13.64}_{\pm 0.15}$	$14.30_{\pm 0.18}$	15.7 (EffNet)	$\mathbf{9.93}_{\pm 0.15}$	$10.31_{\pm 0.15}$	14.5 (DAT)
Food101	$7.02_{\pm 0.02}$	$7.17_{\pm 0.03}$	7.0 (Gpipe)	$\mathbf{3.82}_{\pm 0.01}$	$3.97_{\pm 0.03}$	4.7 (DAT)
ImageNet	$15.14_{\pm 0.03}$	15.3	**14.2** (KDforAA)	$\mathbf{11.39}_{\pm 0.02}$	11.8	11.45 (ViT)

Table 3: Top-1 error rates for finetuning EfficientNet-b7 (left; ImageNet pretraining only) and EfficientNet-L2 (right; pretraining on ImageNet plus additional data, such as JFT) on various downstream tasks. Previous state-of-the-art (SOTA) includes EfficientNet (EffNet) (Tan & Le, 2019), Gpipe (Huang et al., 2018), DAT (Ngiam et al., 2018), BiT-M/L (Kolesnikov et al., 2020), KDforAA (Wei et al., 2020), TBMSL-Net (Zhang et al., 2020), and ViT (Dosovitskiy et al., 2020).

3.3 Robustness to Label Noise

The fact that SAM seeks out model parameters that are robust to perturbations suggests SAM's potential to provide robustness to noise in the training set (which would perturb the training loss landscape). Thus, we assess here the degree of robustness that SAM provides to label noise.

In particular, we measure the effect of applying SAM in the classical noisy-label setting for CIFAR-10, in which a fraction of the training set's labels are randomly flipped; the test set remains unmodified (i.e., clean). To ensure valid comparison to prior work, which often utilizes architectures specialized to the noisy-label setting, we train a simple model of similar size (ResNet-32) for 200 epochs, following Jiang et al. (2019). We evaluate five variants of model training: standard SGD, SGD with Mixup (Zhang et al., 2017), SAM, and "bootstrapped" variants of SGD with Mixup and SAM (wherein the model is first trained as usual and then retrained from scratch on the labels predicted by the initially trained model). When applying SAM, we use $\rho = 0.1$ for all noise levels except 80%, for which we use $\rho = 0.05$ for more stable convergence. For the Mixup baselines, we tried all values of $\alpha \in \{1, 8, 16, 32\}$ and conservatively report the best score for each noise level.

Method	Noise rate (%)			
	20	40	60	80
Sanchez et al. (2019)	94.0	92.8	90.3	74.1
Zhang & Sabuncu (2018)	89.7	87.6	82.7	67.9
Lee et al. (2019)	87.1	81.8	75.4	-
Chen et al. (2019)	89.7	-	-	52.3
Huang et al. (2019)	92.6	90.3	43.4	-
MentorNet (2017)	92.0	91.2	74.2	60.0
Mixup (2017)	94.0	91.5	86.8	76.9
MentorMix (2019)	**95.6**	**94.2**	91.3	**81.0**
SGD	84.8	68.8	48.2	26.2
Mixup	93.0	90.0	83.8	70.2
Bootstrap + Mixup	93.3	92.0	87.6	72.0
SAM	95.1	93.4	90.5	77.9
Bootstrap + SAM	95.4	**94.2**	**91.8**	79.9

Table 4: Test accuracy on the clean test set for models trained on CIFAR-10 with noisy labels. Lower block is our implementation, upper block gives scores from the literature, per Jiang et al. (2019).

As seen in Table 4, SAM provides a high degree of robustness to label noise, on par with that provided by state-of-the art procedures that specifically target learning with noisy labels. Indeed, simply training a model with SAM outperforms all prior methods specifically targeting label noise robustness, with the exception of MentorMix (Jiang et al., 2019). However, simply bootstrapping SAM yields performance comparable to that of MentorMix (which is substantially more complex).

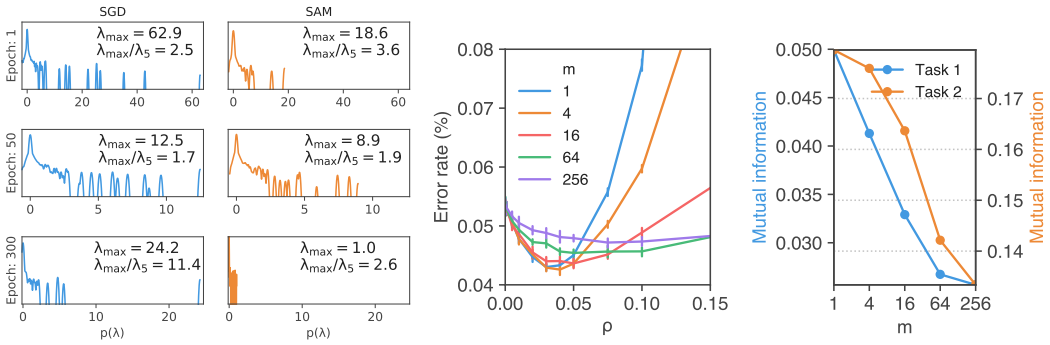

Figure 3: (left) Evolution of the spectrum of the Hessian during training of a model with standard SGD (lefthand column) or SAM (righthand column). (middle) Test error as a function of ρ for different values of m. (right) Predictive power of m-sharpness for the generalization gap, for different values of m (higher means the sharpness measure is more correlated with actual generalization gap).

4 Sharpness and Generalization Through the Lens of SAM

4.1 m-sharpness

Though our derivation of SAM defines the SAM objective over the entire training set, when utilizing SAM in practice, we compute the SAM update per-batch (as described in Algorithm 1) or even by averaging SAM updates computed independently per-accelerator (where each accelerator receives a subset of size m of a batch, as described in Section 3). This latter setting is equivalent to modifying the SAM objective (equation 1) to sum over a set of independent ϵ maximizations, each performed on a sum of per-data-point losses on a disjoint subset of m data points, rather than performing the ϵ maximization over a global sum over the training set (which would be equivalent to setting m to the total training set size). We term the associated measure of sharpness of the loss landscape *m-sharpness*.

To better understand the effect of m on SAM, we train a small ResNet on CIFAR-10 using SAM with a range of values of m. As seen in Figure 3 (middle), smaller values of m tend to yield models having better generalization ability. This relationship fortuitously aligns with the need to parallelize across multiple accelerators in order to scale training for many of today's models.

Intriguingly, the m-sharpness measure described above furthermore exhibits better correlation with models' actual generalization gaps as m decreases, as demonstrated by Figure 3 (right)[7]. In particular, this implies that m-sharpness with $m < n$ yields a better predictor of generalization than the full-training-set measure suggested by Theorem 1 in Section 2 above, suggesting an interesting new avenue of future work for understanding generalization.

4.2 Hessian Spectra

Motivated by the connection between geometry of the loss landscape and generalization, we constructed SAM to seek out minima of the training loss landscape having both low loss value and low curvature (i.e., low sharpness). To further confirm that SAM does in fact find minima having low curvature, we compute the spectrum of the Hessian for a WideResNet40-10 trained on CIFAR-10 for 300 steps both with and without SAM (without batch norm, which tends to obscure interpretation of the Hessian), at different epochs during training. Due to the parameter space's dimensionality, we approximate the Hessian spectrum using the Lanczos algorithm of Ghorbani et al. (2019).

Figure 3 (left) reports the resulting Hessian spectra. As expected, the models trained with SAM converge to minima having lower curvature, as seen in the overall distribution of eigenvalues, the

[7]We follow the rigorous framework of Jiang et al. (2019), reporting the mutual information between the m-sharpness measure and generalization on the two publicly available tasks from the *Predicting generalization in deep learning* NeurIPS2020 competition. https://competitions.codalab.org/competitions/25301

maximum eigenvalue ($\lambda_{\max}$) at convergence (approximately 24 without SAM, 1.0 with SAM), and the bulk of the spectrum (the ratio $\lambda_{\max}/\lambda_5$, commonly used as a proxy for sharpness (Jastrzebski et al., 2020); up to 11.4 without SAM, and 2.6 with SAM).

5 Related Work

The idea of searching for "flat" minima can be traced back to Hochreiter & Schmidhuber (1995), and its connection to generalization has seen significant study (Shirish Keskar et al., 2016; Dziugaite & Roy, 2017; Neyshabur et al., 2017; Dinh et al., 2017). In a recent large scale empirical study, Jiang et al. (2019) studied 40 complexity measures and showed that a sharpness-based measure has highest correlation with generalization, which motivates penalizing sharpness. Hochreiter & Schmidhuber (1997) was perhaps the first paper on penalizing the sharpness, regularizing a notion related to Minimum Description Length (MDL). Other ideas which also penalize sharp minima include operating on diffused loss landscape (Mobahi, 2016) and regularizing local entropy (Chaudhari et al., 2016). Another direction is to not penalize the sharpness explicitly, but rather average weights during training; Izmailov et al. (2018) showed that doing so can yield flatter minima that can also generalize better. However, the measures of sharpness proposed previously are difficult to compute and differentiate through. In contrast, SAM is highly scalable as it only needs two gradient computations per iteration. The concurrent work of Sun et al. (2020) focuses on resilience to random and adversarial corruption to expose a model's vulnerabilities; this work is perhaps closest to ours. Our work has a different basis: we develop SAM motivated by a principled starting point in generalization, clearly demonstrate SAM's efficacy via rigorous large-scale empirical evaluation, and surface important practical and theoretical facets of the procedure (e.g., m-sharpness). The notion of all-layer margin introduced by Wei & Ma (2020) is closely related to this work; one is adversarial perturbation over the activations of a network and the other over its weights, and there is some coupling between these two quantities.

6 Discussion and Future Work

In this work, we have introduced SAM, a novel algorithm that improves generalization by simultaneously minimizing loss value and loss sharpness; we have demonstrated SAM's efficacy through a rigorous large-scale empirical evaluation. We have surfaced a number of interesting avenues for future work. On the theoretical side, the notion of per-data-point sharpness yielded by m-sharpness (in contrast to global sharpness computed over the entire training set, as has typically been studied in the past) suggests an interesting new lens through which to study generalization. Methodologically, our results suggest that SAM could potentially be used in place of Mixup in robust or semi-supervised methods that currently rely on Mixup (giving, for instance, MentorSAM). We leave to future work a more in-depth investigation of these possibilities.

7 Acknowledgments

We thank our colleagues at Google — Atish Agarwala, Xavier Garcia, Dustin Tran, Yiding Jiang, Basil Mustafa, Samy Bengio — for their feedback and insightful discussions. We also thank the JAX and FLAX teams for going above and beyond to support our implementation. We are grateful to Sven Gowal for his help in replicating EfficientNet using JAX, and Justin Gilmer for his implementation of the Lanczos algorithm[8] used to generate the Hessian spectra. We thank Niru Maheswaranathan for his matplotlib mastery. We also thank David Samuel for providing a PyTorch implementation of SAM[9].

References

Naman Agarwal, Rohan Anil, Elad Hazan, Tomer Koren, and Cyril Zhang. Revisiting the generalization of adaptive gradient methods, 2020. URL https://openreview.net/forum?id=BJl6t64tvr.

[8] https://github.com/google/spectral-density

[9] https://github.com/davda54/sam